

Tranexamic Acid (TXA)

(Cyklokapron)

I. Classification

- Antifibrinolytic Agent, topical Hemostatic agent

II. Action(s)

- Forms a reversible complex that displaces plasminogen from fibrin that inhibits fibrinolysis. Inhibits the proteolytic activity of plasmin thus **decreasing the breakdown of already formed clot structures.**

III. Indication(s)

- Hypovolemic shock caused by massive traumatic hemorrhage
Trauma less than 3 hours of injury with poor peripheral circulation, BP<90 mmHg or Pulse > 110 or both. If Age > 65 BP<110 or Pulse > 100 (CRASH-2 Trial Collaborators 2010; CRASH-2 Trial Collaborators 2011)
- Potential use with shock caused from postpartum or GI bleed (Medical Director Consult)

IV. Contraindications

- Only use with serious Hemorrhagic SHOCK

V. Drug interactions

- Should be administered in separate IV from other medications if possible and do not administer in same IV line as blood products

VI. Adult Dosing and Administration

- 2000 mg (2 gram) IV/IO slow push

VII. Pediatric Dosing and Administration

- 10 mg/kg (max 2g)

VIII. Onset

- Immediate

IX. Duration

- Half Life 2 hours (excreted in urine unchanged)

X. Possible Adverse Effects

- Headache
- Worsening cerebral edema if used in patients with Subarachnoid Hemorrhage
- CNS depression and/or Seizure
- Thrombotic complications such as DVT, PE, retinal vein/artery occlusion (rare)